HYPERCOVERING DESCENT FOR SHEAVES OF SETS AND ABELIAN GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-hypercovering-descent-sheaves names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof carries the abelian case by reduction to sheaves of sets ('a^-1 commutes with products'), carries the adjunction half in full, and carries the essential-image half through the descent-data argument; the single 'small detail omitted' (faithfulness of pullback along a covering) is a one-step standard fact inside a complete architecture - recorded honestly here per the criterion-very-ample-bis precedent.

Allowed standard prerequisites (editorial).

- Cartesian objects under the augmentation. Same chapter, \label{lemma-augmentation-cartesian-module}. Line 4191: a^-1F is cartesian for F in Sh(C).
- a_*a^-1 as an equalizer. Same chapter, \label{lemma-comparison}. Line 4204: identifies a_*a^-1F with the equalizer of the two face pushforwards.
- Push-pull on localized sites. Same chapter, \label{lemma-push-pull-localization}. Lines 4206-4210: rewrites a_{i,*}a_i^-1F as SheafHom(F(K_i)^#, F).
- Cartesian sheaves characterized by descent data. Same chapter, \label{lemma-characterize-cartesian}. Lines 4253-4256 and 4302-4305: reformulates the problem as descent data on K_1/K_2.
- Localization comparison for the total site. Same chapter, \label{lemma-localize-compare}. Line 4255: describes Sh(C/K_n), letting the problem be posed on C alone.
- Good sites for a topos. Sites chapter, \label{sites-lemma-topos-good-site}. Lines 4274-4279: replaces C by a site with all finite limits and a simplicial representable K.
- Face and degeneracy maps of a simplicial object. Simplicial chapter, \label{simplicial-definition-face-degeneracy}. Lines 4292-4294: notation d/s for the maps of U.

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- Work-out of the cosk_1 terms. Simplicial chapter, \label{simplicial-lemma-work-out}. Line 4310: describes the image of (d_0^2,d_1^2,d_2^2) on T-valued points.
- Glueing maps along a covering. Sites chapter, \label{sites-lemma-glue-maps}. Lines 4330-4334: descends alpha to an isomorphism pr_1^-1G_0 -> pr_2^-1G_0.
- Mapping property of glueing. Sites chapter, \label{sites-lemma-mapping-property-glue}. Lines 4347-4348: finishes by glueing over the covering {U_0 -> X}.

Source reference. Original lemma 0D7H: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2530–2530. Prerequisite pointer invoked at proof line(s) 4191; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09WM: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3983–3983. Prerequisite pointer invoked at proof line(s) 4204; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D86: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3708–3708. Prerequisite pointer invoked at proof line(s) 4206; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D7I: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2561–2561. Prerequisite pointer invoked at proof line(s) 4253; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D87: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 3736–3736. Prerequisite pointer invoked at proof line(s) 4255; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 04TP: sources/stacks-chow/tags.txt, source lines 6262–6262. Prerequisite pointer invoked at proof line(s) 4212; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03CI: sources/stacks-chow/tags.txt, source lines 4385–4385. Prerequisite pointer invoked at proof line(s) 4274; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0165: sources/stacks-chow/tags.txt, source lines 1532–1532. Prerequisite pointer invoked at proof line(s) 4292; it names a

standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0187: sources/stacks-chow/tags.txt, source lines 1606–1606. Prerequisite pointer invoked at proof line(s) 4310; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04TQ: sources/stacks-chow/tags.txt, source lines 6263–6263. Prerequisite pointer invoked at proof line(s) 4330; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04TS: sources/stacks-chow/tags.txt, source lines 6265–6265. Prerequisite pointer invoked at proof line(s) 4347; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. Then*

(1) *$a^{-1} : Sh(\mathcal{C}) \to Sh((\mathcal{C}/K)_{total})$ is fully faithful with essential image the cartesian sheaves of sets,*
(2) *$a^{-1} : Ab(\mathcal{C}) \to Ab((\mathcal{C}/K)_{total})$ is fully faithful with essential image the cartesian sheaves of abelian groups.*

In both cases a_* provides the quasi-inverse functor.

Proof. The case of abelian sheaves follows immediately from the case of sheaves of sets as the functor a^{-1} commutes with products. In the rest of the proof we work with sheaves of sets. Observe that $a^{-1}\mathcal{F}$ is cartesian for $\mathcal{F}$ in $Sh(\mathcal{C})$ by Lemma 0D7H. It suffices to show that the adjunction map $\mathcal{F} \to a_*a^{-1}\mathcal{F}$ is an isomorphism $\mathcal{F}$ in $Sh(\mathcal{C})$ and that for a cartesian sheaf $\mathcal{G}$ on $(\mathcal{C}/K)_{total}$ the adjunction map $a^{-1}a_*\mathcal{G} \to \mathcal{G}$ is an isomorphism.

Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. Recall that $a_*a^{-1}\mathcal{F}$ is the equalizer of the two maps $a_{0,*}a_0^{-1}\mathcal{F} \to a_{1,*}a_1^{-1}\mathcal{F}$, see Lemma 09WM. By Lemma 0D86

$$a_{0,*}a_0^{-1}\mathcal{F} = \mathcal{H}om(F(K_0)^\#, \mathcal{F}) \quad \text{and} \quad a_{1,*}a_1^{-1}\mathcal{F} = \mathcal{H}om(F(K_1)^\#, \mathcal{F})$$

On the other hand, we know that

$$F(K_1)^\# \rightrightarrows F(K_0)^\# \longrightarrow \text{final object } * \text{ of } Sh(\mathcal{C})$$

is a coequalizer diagram in sheaves of sets by definition of a hypercovering. Thus it suffices to prove that $\mathcal{H}om(-, \mathcal{F})$ transforms coequalizers into equalizers which is immediate from the construction in Sites, Section 04TP.

Let $\mathcal{G}$ be a cartesian sheaf on $(\mathcal{C}/K)_{total}$. We will show that $\mathcal{G} = a^{-1}\mathcal{F}$ for some sheaf $\mathcal{F}$ on $\mathcal{C}$. This will finish the proof because then $a^{-1}a_*\mathcal{G} = a^{-1}a_*a^{-1}\mathcal{F} = a^{-1}\mathcal{F} = \mathcal{G}$ by the result of the previous paragraph. Set $\mathcal{K}_n = F(K_n)^\#$ for $n \geq 0$. Then we have maps of sheaves

$$\mathcal{K}_2 \rightrightarrows \mathcal{K}_1 \rightrightarrows \mathcal{K}_0$$

coming from the fact that K is a simplicial semi-representable object. The fact that K is a hypercovering means that

$$\mathcal{K}_1 \to \mathcal{K}_0 \times \mathcal{K}_0 \quad \text{and} \quad \mathcal{K}_2 \to \Big(\text{cosk}_1(\ \mathcal{K}_1 \rightrightarrows \mathcal{K}_0\ )\Big)_2$$

are surjective maps of sheaves. Using the description of cartesian sheaves on $(\mathcal{C}/K)_{total}$ given in Lemma 0D7I and using the description of $Sh(\mathcal{C}/K_n)$ in Lemma 0D87 we find that our problem can be entirely formulated[1] in terms of

(1) the topos $Sh(\mathcal{C})$, and
(2) the simplicial object $\mathcal{K}$ in $Sh(\mathcal{C})$ whose terms are $\mathcal{K}_n$.

Thus, after replacing $\mathcal{C}$ by a different site $\mathcal{C}'$ as in Sites, Lemma 03CI, we may assume $\mathcal{C}$ has all finite limits, the topology on $\mathcal{C}$ is subcanonical, a family $\{V_j \to V\}$ of morphisms of $\mathcal{C}$ is a covering if and only if $\coprod h_{V_j} \to V$ is surjective, and there exists a simplicial object U of $\mathcal{C}$ such that $\mathcal{K}_n = h_{U_n}$ as simplicial sheaves. Working backwards through the equivalences we may assume $K_n = \{U_n\}$ for all n.

Let X be the final object of $\mathcal{C}$. Then $\{U_0 \to X\}$ is a covering, $\{U_1 \to U_0 \times U_0\}$ is a covering, and $\{U_2 \to (\text{cosk}_1\text{sk}_1 U)_2\}$ is a covering. Let us use $d_i^n : U_n \to U_{n-1}$ and $s_j^n : U_n \to U_{n+1}$ the morphisms corresponding to δ_i^n and σ_j^n as in Simplicial, Definition 0165. By abuse of notation, given a morphism $c : V \to W$ of $\mathcal{C}$ we denote the morphism of topoi $c : Sh(\mathcal{C}/V) \to Sh(\mathcal{C}/W)$ by the same letter. Now $\mathcal{G}$ is given by a sheaf $\mathcal{G}_0$ on $\mathcal{C}/U_0$ and an isomorphism $\alpha : (d_1^1)^{-1}\mathcal{G}_0 \to (d_0^1)^{-1}\mathcal{G}_0$ satisfying the cocycle condition on $\mathcal{C}/U_2$ formulated in Lemma 0D7I. Since $\{U_2 \to (\text{cosk}_1\text{sk}_1 U)_2\}$ is a covering, the corresponding pullback functor on sheaves is faithful (small detail omitted). Hence we may replace U by $\text{cosk}_1\text{sk}_1 U$, because this replaces U_2 by $(\text{cosk}_1\text{sk}_1 U)_2$ and leaves U_1 and U_0 unchanged. Then

$$(d_0^2, d_1^2, d_2^2) : U_2 \to U_1 \times U_1 \times U_1$$

is a monomorphism whose its image on T-valued points is described in Simplicial, Lemma 0187. In particular, there is a morphism c fitting into a commutative diagram

$$\begin{array}{ccc} U_1 \times_{(d_1^1, d_0^1), U_0 \times U_0, (d_1^1, d_0^1)} U_1 & \xrightarrow[c]{} & U_2 \\ \downarrow & & \downarrow \\ U_1 \times U_1 & \xrightarrow{(\text{pr}_1, \text{pr}_2, s_0^0 \circ d_1^1 \circ \text{pr}_1)} & U_1 \times U_1 \times U_1 \end{array}$$

as going around the other way defines a point of U_2. Pulling back the cocycle condition for α on U_2 translates into the condition that the pullbacks of α via the projections to $U_1 \times_{(d_1^1, d_0^1), U_0 \times U_0, (d_1^1, d_0^1)} U_1$ are the same as the pullback of α via $s_0^0 \circ d_1^1 \circ \text{pr}_1$ is the identity map (namely, the pullback of α by s_0^0 is the identity). By Sites, Lemma 04TQ this means that α comes from an isomorphism

$$\alpha' : \text{pr}_1^{-1}\mathcal{G}_0 \to \text{pr}_2^{-1}\mathcal{G}_0$$

of sheaves on $\mathcal{C}/U_0 \times U_0$. Then finally, the morphism $U_2 \to U_0 \times U_0 \times U_0$ is surjective on associated sheaves as is easily seen using the surjectivity of $U_1 \to U_0 \times U_0$ and

[1] Even though it does not matter what the precise formulation is, we spell it out: the problem is to show that given an object $\mathcal{G}_0/\mathcal{K}_0$ of $Sh(\mathcal{C})/\mathcal{K}_0$ and an isomorphism

$$\alpha : \mathcal{G}_0 \times_{\mathcal{K}_0, \mathcal{K}(\delta_1^1)} \mathcal{K}_1 \to \mathcal{G}_0 \times_{\mathcal{K}_0, \mathcal{K}(\delta_0^1)} \mathcal{K}_1$$

over $\mathcal{K}_1$ satisfying a cocycle condition in $Sh(\mathcal{C})/\mathcal{K}_2$, there exists $\mathcal{F}$ in $Sh(\mathcal{C})$ and an isomorphism $\mathcal{F} \times \mathcal{K}_0 \to \mathcal{G}_0$ over $\mathcal{K}_0$ compatible with α.

the description of U_2 given above. Therefore α' satisfies the cocycle condition on $U_0 \times U_0 \times U_0$. The proof is finished by an application of Sites, Lemma 04TS to the covering $\{U_0 \to X\}$. $\square$